



Girijananda Chowdhury University Guwahati 781017, Assam

UNIT TEST - I

BCE23405T Structural Analysis

Session: 2024-2025

Batch: 2023-2027

Date :: 26 February 2025

Time : 1 hour 30 Min

Total Marks: 30

Solve Any THREE questions

Question 1 (10 marks)

[CO1]

Determine the support reactions and the forces in the members AD, CF, DF and CE using method of sections given in Figure 1.

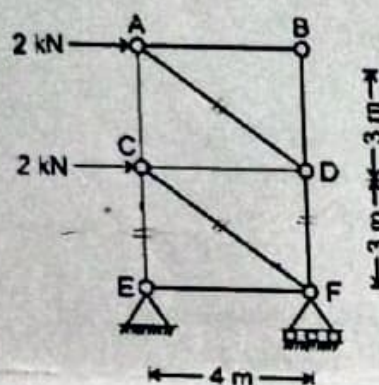


Figure 1

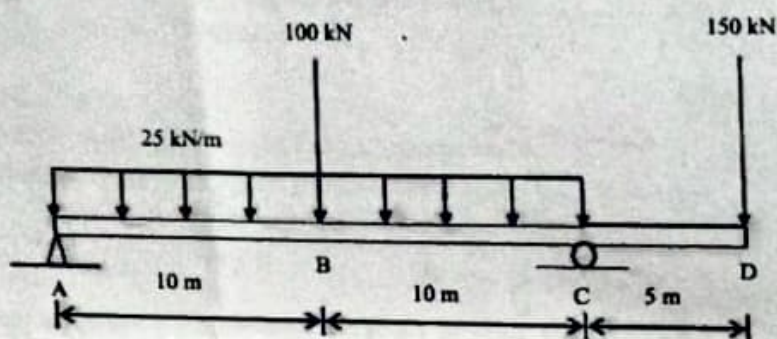


Figure 2

Question 2 (10 marks)

[CO1]

Consider a simply supported beam ABCD (Figure 2). By taking E is 200 GPa and I is $500 \times 10^6 \text{ mm}^4$. (i) Show that the beam is statically determinate beam. (ii) Using Unit Load Method determine vertical displacement at joint D.

Question 3 (10 marks)

[CO1]

Determine the vertical deflection at Joint C (Figure 3) using Castigliano's Theorem. Take $E = 200 \text{ GPa}$ and $I = 550 \times 10^6 \text{ mm}^4$.

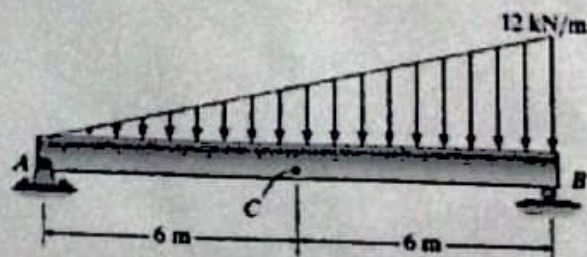


Figure 3

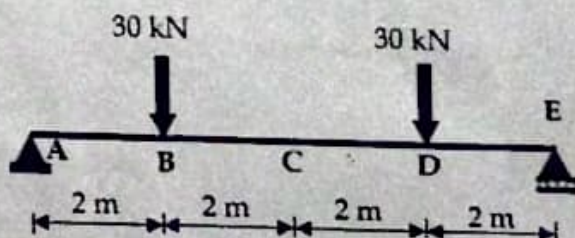


Figure 4

Question 4 (10 marks)

[CO1]

Determine the vertical deflection at Joint C using Castigliano's Theorem (Figure 4). Take $E = 200 \text{ GPa}$ and $I = 600 \times 10^6 \text{ mm}^4$.